

Section AQuestion 2

$$) (101101)_2$$

$$\begin{array}{cccccc} 1 & 0 & 1 & 1 & 0 & 1 \\ 5 & 4 & 3 & 2 & 1 & 0 \end{array}$$

$$\begin{aligned} &= 2^5 + 2^3 + 2^2 + 2^0 \\ &= 32 + 8 + 4 + 1 \\ &= 45 \end{aligned}$$

45.5
50

Good!

Amithan
3/07/26

$$) (1110101)_2$$

$$\begin{array}{ccccccc} 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 6 & 5 & 4 & 3 & 2 & 1 & 0 \end{array}$$

$$\begin{aligned} &= 2^6 + 2^5 + 2^4 + 2^2 + 2^0 \\ &= 64 + 32 + 16 + 4 + 1 \\ &= 117 \end{aligned}$$

Rem: Try to avoid careless mistakes.

Explain your answers briefly.

It helps the teacher to check the paper & give the required marks.

$$) (110101.101)_2$$

$$\begin{array}{ccccccc} 1 & 1 & 0 & 1 & 0 & 1 & . & 1 & 0 & 1 \\ 5 & 4 & 3 & 2 & 1 & 0 & & -1 & -2 & -3 \end{array}$$

$$2^5 + 2^4 + 2^2 + 2^0 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}$$

$$32 + 16 + 4 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$53 + \frac{7}{8}$$

$$53 + \frac{7}{8} = 53.875$$

5

d) $(10011.011)_2$

$$\begin{array}{ccccccc} 1 & 0 & 0 & 1 & 1 & . & 0 & 1 & 1 \\ 4 & 3 & 2 & 1 & 0 & & -1 & -2 & -3 \end{array}$$

$$2^4 + 2^3 + 2^2 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}$$

$$16 + 8 + 4 + \frac{4}{8} + \frac{2}{8} + \frac{1}{8}$$

$$19 + \frac{7}{8}$$

$$= 19.875$$

2.5

Question 2

a)

$$\begin{array}{r|l} 2 & 57 \\ \hline 2 & 28 \quad 1 \\ 2 & 14 \quad 0 \\ 2 & 7 \quad 0 \\ 2 & 3 \quad 1 \\ & 1 \quad 1 \end{array}$$

Ans = 111001

1)

~~101~~

$$2 \overline{) 125}$$

$$2 \overline{) 62} \quad 1$$

$$2 \overline{) 31} \quad 0$$

$$2 \overline{) 15} \quad 1$$

$$2 \overline{) 7} \quad 1$$

$$2 \overline{) 3} \quad 1$$

$$\text{Ans} = \underline{1111101}$$

c)

$$2 \overline{) 45}$$

$$2 \overline{) 22} \quad 1$$

$$2 \overline{) 11} \quad 0$$

$$2 \overline{) 5} \quad 1$$

$$2 \overline{) 2} \quad 1$$

$$\underline{\underline{101101.101}}$$

d)

$$2 \overline{) 23}$$

$$2 \overline{) 11} \quad 1$$

$$2 \overline{) 5} \quad 1$$

$$2 \overline{) 2} \quad 1$$

$$\underline{\underline{10111.011}}$$

14

$$2 \times 6.625 \times 2$$

$$2 \times 1 \times 2.5 \times 2$$

$$2 \times 0 \times 5 \times 2$$

$$2 \times 1 \times 0$$

$$2 \times 0 \times 0.375$$

$$2 \times 0.1 \times 0.75$$

$$2 \times 0.1 \times 0.5$$

$$0.0$$

Question 3

a) $(11011101)_2$

01	000	0
01	001	1
10	010	2
10	011	3
01	100	4
01	101	5
00	110	6
00	111	7

675

b) $(572)_{10}$

$= (10111010)_2$

c)

~~00000000~~

~~11~~

c) $(101101011)_2$

$(16)_{10}$

16

0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

$$(9AF)_{16} = (100110101111)_2$$

$$\begin{array}{r} 2 \overline{) 156} \\ \underline{2 \overline{) 28}} \\ 2 \overline{) 39} \\ \underline{2 \overline{) 19}} \\ 2 \overline{) 9} \\ \underline{2 \overline{) 5}} \\ 2 \overline{) 2} \\ \underline{1} \end{array} \quad \begin{array}{r} 16 \overline{) 156} \\ \underline{16 \overline{) 9}} \\ 9 \end{array}$$

$$(156)_{10} = (9C)_{16}$$

$$\begin{array}{r} 1 \\ 0 \end{array}$$

$$= (10011100)_2$$

$$= (9C)_{16}$$

$$3 \times 16^2 + 14 \times 16^1 + 9 \times 16^0$$

$$(3E9)_{16} = 111101001 \quad 3 \times 256 + 224 + 9$$

$$\begin{array}{r} 1 1 1 0 0 0 1 \\ 9 8 6 5 3 1 \end{array} = 1001$$

$$768 + 233$$

$$= 2^9 + 2^8 + 2^7 + 2^6 + 2^3 + 2^0$$

$$= 512 + 256 + 128 + 64 + 8 + 1 + 32 = 969$$

$$= 968 + 33 = 1001$$

5

Questions

$$(a_2) \quad (5 \text{ DF}) = \frac{(100 \cdot 110 \cdot 111)}{2}$$

$$2 \times 10^3 (2327) 8$$

$$(68) \quad (745)_8 = (\underline{11110010})_2 = (1E5)_{16}$$

c) $(A_3 F)_8 = (1010001111)_2 = (F077)_8$

Section B

Section 5

1	0	1	1	0	1
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$$\begin{array}{r} 1101011 \\ 1101111 \\ \hline \end{array}$$

6	110101
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1	1	0	1	0	1	1
1	0	1	0	0	1	1

Questions

~~181101 = ~~010010~~~~

$$\begin{array}{r} 100100 \\ 110110 \\ \hline 011011 \end{array}$$

	101101	(9)
	101101	(10)

$$\begin{array}{r} 011011 \\ \hline 100100 \\ \hline 1010001 \end{array} \quad (1's \text{ complement})$$

$$\begin{array}{r} 10016 \\ \times 71 \\ \hline \end{array}$$

no showing the steps

(2's complement)

$$\begin{array}{r} 1111000 = 1111000 \\ 0110101 = 1001011 \\ \hline + 1000011 \end{array}$$

Question 3

(1)

$$\begin{array}{r} 1011 \\ \hline 110 \\ 0 \end{array}$$

$$\begin{array}{r} 1011x \\ \hline 1011xx \\ \hline 1000010 \end{array}$$

$$\begin{array}{r} 1100 \\ 10111100 \end{array}$$

(6)

$$\begin{array}{r} 101 \\ \hline 161 \\ \hline -101 \\ 000 \end{array}$$

$$\begin{array}{l} \text{Quotient} = 1100 \\ \text{Remainder} = 0 \end{array}$$

27)

Lehon CQuestion 8

a)

$$\begin{array}{r} 10 \\ 4 \overline{) 7} \\ \underline{3} \\ 13 \end{array}$$

$$\begin{array}{r} 3 \\ \underline{3} \\ 0 \end{array}$$

b)

$$\begin{array}{r} 10 \\ 8 \overline{) 4} \\ \underline{8} \\ 4 \end{array}$$

$$\begin{array}{r} 5 \\ \underline{5} \\ 0 \end{array}$$

$$\begin{array}{r} 6 \\ \underline{6} \\ 0 \end{array}$$

c)

$$\begin{array}{r} 10 \\ 2 \overline{) 4} \\ \underline{2} \\ 2 \end{array}$$

$$\begin{array}{r} 4 \\ \underline{4} \\ 0 \end{array}$$

$$\begin{array}{r} 8 \\ \underline{8} \\ 0 \end{array}$$

d)

$$\begin{array}{r} 3 \\ 3 \overline{) 9} \\ \underline{9} \\ 0 \end{array}$$

$$\begin{array}{r} 12 \\ \underline{12} \\ 0 \end{array}$$

$$\begin{array}{r} 0 \\ \underline{0} \\ 0 \end{array}$$

$$\begin{array}{r} 0 \\ \underline{0} \\ 0 \end{array}$$

$$\begin{array}{r} 34 \\ \underline{34} \\ 0 \end{array}$$

$$\begin{array}{r} 0 \\ \underline{0} \\ 0 \end{array}$$

8

Question 9

$$\begin{array}{r}
 657 \\
 2 \overline{) 1124} \\
 \underline{11} \\
 0 \\
 \underline{0} \\
 0 \\
 \underline{0} \\
 0
 \end{array}$$

Ans = 11248 ✓

$$\begin{array}{r}
 622 \\
 5 \overline{) 3110} \\
 \underline{30} \\
 10 \\
 \underline{10} \\
 0 \\
 \underline{0} \\
 0
 \end{array}$$

$$\begin{array}{r}
 37 \\
 5 \overline{) 233} \\
 \underline{20} \\
 33 \\
 \underline{30} \\
 3
 \end{array}$$

$$\begin{array}{r}
 122 \\
 6 \overline{) 759} \\
 \underline{6} \\
 15 \\
 \underline{12} \\
 39 \\
 \underline{36} \\
 3
 \end{array}$$

quotient = 122
Remainder = 3

⑧